

■ G-STORE RAJAM

Model Context Protocol (MCP) Server — Complete 23 Functions Reference

Comprehensive Manual & Operational Guide for Google Gemini Spark & AWS AppSync

Target Store: G-Store (Rice & Kirana Mart), Rajam, A.P.	Cloud Backend: AWS AppSync GraphQL & DynamoDB
AI Interface: Google Gemini Spark (MCP Standard 2024-11-05)	Media Storage: Cloudinary CDN (ricemart_products)
Total Tools: 23 Active Functions	Status: Production Ready & Tested

1. Catalog & Multi-Variant Management (11 Tools)

Enables store managers to create products, update details, generate 600x600 AI packaging photos, manage multi-size pack variants, and restock wholesale batches.

1.1 list_products

Lists all catalog products with pack sizes, selling prices, MRPs, and stock status. Supports category filtering and listing status toggles.

Parameter	Type	Req	Description
category	string	No	Filter by 'Rice Bags', 'Cooking Oils', 'Dals & Pulses', 'Dairy Essentials', 'Spices & Masalas'
includeUnlisted	boolean	No	If true, includes hidden/unlisted products (default: false)
Example Gemini Prompt:	"Gemini, list all products under Rice Bags with their current prices and stock counts."		
Result Output:	Returns count and array of products with formatted variants: 26kg bag @ ■1450 (MRP ■1600, Stock: 45).		

1.2 get_product_by_id

Retrieves comprehensive information, all pack size variants, Cloudinary image URLs, and English/Telugu descriptions for a specific product ID.

Parameter	Type	Req	Description
productId	string	Yes	The unique product ID (e.g. 'p_rice_sona', 'p_oil_freedom')
Example Gemini Prompt:	"Gemini, show me full product details for product ID p_rice_sona."		
Result Output:	Returns product name, Telugu name, Cloudinary image URL, and all available pack variants.		

1.3 create_product

Creates a new product in the G-Store catalog, automatically uploading the image to Cloudinary folder 'ricemart_products' and linking it to AWS AppSync.

Parameter	Type	Req	Description
name	string	Yes	English display name (e.g. 'Amul Taaza Toned Milk')
category	string	Yes	Category: 'Dairy Essentials', 'Rice Bags', 'Cooking Oils', etc.
price	number	Yes	Selling price in ₹ (e.g. 34)
mrp	number	No	Printed MRP in ₹ (defaults to ~15% above selling price)
size	string	No	Weight/volume size (e.g. '500ml', '1L', '26kg', '10kg')
stock	number	No	Initial stock count in units/bags (default: 50)
description	string	No	English description or brand notes
imageUrl	string	No	Source image URL or base64 (auto-uploaded to Cloudinary)

Example Gemini Prompt:	"Gemini, add a new product: Fortune Sunlite Refined Sunflower Oil 1L at ₹125 (MRP 140), stock 40 packets, under Cooking Oils."
Result Output:	Created product p_oil_fortune_sunlite with Cloudinary CDN URL and formatted variant in DynamoDB.

1.4 update_product_details

Updates product names, Telugu names, brand, category, description, image, or listing visibility without deleting or recreating the item.

Parameter	Type	Req	Description
productId	string	Yes	The unique product ID to modify
name	string	No	New English display name
nameTe	string	No	New Telugu display name (e.g. 'పిల్లల పులి, పిల్లల పులి పుడికల') (max 100 characters)
brand	string	No	Brand name (e.g. 'Lalitha', 'Heritage', 'Freedom')
category	string	No	New category key
description	string	No	New English description
descriptionTe	string	No	New Telugu description
imageUrl	string	No	New image URL (auto-uploaded to Cloudinary)
isListed	boolean	No	Set to false to hide item from customer app

Example Gemini Prompt:	"Gemini, update product p_rice_lalitha: set Telugu name to పిల్లల పులి, పిల్లల పులి పుడికల and brand to Lalitha."
Result Output:	Product details updated successfully in AppSync with Telugu name and brand tags.

1.5 generate_and_upload_product_image

Generates an 8K studio packaging photo on a clean white background using AI, or processes a reference photo taken on a phone, automatically applying 600x600 square padding and uploading to Cloudinary CDN.

Parameter	Type	Req	Description
productName	string	Yes	Product name (e.g. 'Lalitha Brand HMT Rice 26kg')
productId	string	No	Optional product ID to immediately link the image in AWS
referenceImageUrlOrBase64	string	No	Optional phone photo URL or base64 image data
customPrompt	string	No	Optional visual details (e.g. 'yellow woven bag with red logo')
category	string	No	Product category preset

Example Gemini Prompt:	"Gemini, generate a clean studio packaging image for Lalitha Brand HMT Rice 26kg bag and attach it to p_rice_lalitha."
Result Output:	Generated 600x600 padded image on white background, uploaded to Cloudinary, attached to product.

1.6 add_product_variant

Adds a new weight/volume size variant (e.g. adding 10kg alongside an existing 26kg bag) with independent price, MRP, and stock quantity under the same listing.

Parameter	Type	Req	Description
productId	string	Yes	Product ID to receive the new variant
size	string	Yes	Size string (e.g. '10kg', '5kg', '500ml', '1L', '200g')
price	number	Yes	Selling price in ₹ for this size
mrp	number	No	Printed MRP in ₹
stock	number	No	Initial stock count (default: 50)

Example Gemini Prompt:	"Gemini, add a 10kg variant at ₹580 (MRP 650) to Sona Masoori Rice (p_rice_sona) with 30 bags stock."
Result Output:	Added variant 10kg to p_rice_sona. App displays 10kg option in size selector.

1.7 delete_product_variant

Removes a specific size variant (e.g. deleting the 5kg option) while keeping the product listing and all other sizes intact.

Parameter	Type	Req	Description
productId	string	Yes	Product ID
variantSize	string	Yes	Size to remove (e.g. '5kg', '500ml')

Example Gemini Prompt:	"Gemini, remove the 5kg size variant from Sona Masoori Rice (p_rice_sona)."
Result Output:	Variant 5kg removed successfully. Product now offers 10kg and 26kg sizes.

1.8 update_product_price_or_stock

Updates the selling price or stock quantity for an existing variant size in a single instant operation.

Parameter	Type	Req	Description
productId	string	Yes	Product ID
variantSize	string	No	Target variant size (e.g. '26kg')
newPrice	number	No	New selling price in ■
stock	number	No	New absolute stock quantity

Example Gemini Prompt:	"Gemini, change the price of Lalitha HMT Rice 26kg to ■1480 and set stock to 45."
Result Output:	Variant price updated to ■1480, stock set to 45 bags.

1.9 bulk_update_stock_or_prices

Batch updates stock levels (setting absolute count or adding stock increments) and prices across multiple products in one atomic call. Ideal for wholesale truck arrivals.

Parameter	Type	Req	Description
updates	array	Yes	List of { productId, variantSize?, newPrice?, stock?, addStock? }

Example Gemini Prompt:	"Gemini, a wholesale truck arrived: add 50 bags to Lalitha HMT (p_rice_lalitha) and add 40 bags to BPT Rice (p_rice_bpt)."
Result Output:	Updated 2 products in batch. Stock incremented by +50 and +40 bags.

1.10 delete_product

Permanently deletes a product and all of its variants from the store catalog.

Parameter	Type	Req	Description
productId	string	Yes	ID of the product to delete

Example Gemini Prompt:	"Gemini, delete product p_rice_old_discontinued from the catalog."
Result Output:	Product deleted successfully from AppSync DynamoDB table.

1.11 get_low_stock_alerts

Identifies and ranks all products running low on inventory below a given threshold units/bags so the store owner knows what to reorder.

Parameter	Type	Req	Description
threshold	number	No	Stock threshold to trigger alert (default: 10 units)

Example Gemini Prompt:	"Gemini, show me all items running low on stock below 15 units."
Result Output:	Found 4 low-stock items (e.g. Heritage Milk: 3 packets left, Freedom Oil: 6 bottles left).

2. Orders & Omnichannel Fulfillment (5 Tools)

Enables recording phone/walk-in orders, auto-deducting inventory, generating printable WhatsApp invoices, and tracking delivery status.

2.1 create_order

Records counter, phone, or WhatsApp customer orders directly into DynamoDB/AppSync. Automatically verifies current catalog prices, computes totals, and deducts inventory stock.

Parameter	Type	Req	Description
customerName	string	Yes	Customer full name (e.g. 'Ramesh', 'Sita Devi')
customerPhone	string	Yes	Customer 10-digit mobile number
deliveryAddress	string	Yes	House no, landmark, or street in Rajam
items	array	Yes	List of { productIdOrName, variantSize?, quantity, price? }
status	string	No	Initial status: 'PENDING', 'PREPARING', 'OUT_FOR_DELIVERY', 'DELIVERED' (default: PENDING)
deliveryFee	number	No	Delivery fee in ₹ (default: 0)

Example Gemini Prompt:

"Gemini, create a COD order for Ramesh (Phone: 9876543210) at D.No 4-12, Near SBI Bank, Rajam: 1 bag Lalitha HMT Rice 26kg and 2 packets Freedom Sunflower Oil 1L."

Result Output:

Order #G-1786896453001-823 recorded. Total: ₹1,690. Stock auto-deducted.

2.2 generate_order_invoice

Generates a clean, professional text receipt and packing slip formatted for printing or copying directly into WhatsApp for customers and delivery boys.

Parameter	Type	Req	Description
orderId	string	Yes	The unique order ID (e.g. 'G-1786896453001-823')

Example Gemini Prompt:

"Gemini, generate a WhatsApp receipt and bill for order G-1786896453001-823."

Result Output:

Generates formatted bill: Store name, Order ID, Customer, Address, Itemized totals, Delivery fee, and Total due.

2.3 list_orders

Fetches recent customer orders from AWS AppSync with status filtering (PENDING, PREPARING, OUT_FOR_DELIVERY, DELIVERED, CANCELLED).

Parameter	Type	Req	Description
status	string	No	Filter: 'PENDING', 'PREPARING', 'OUT_FOR_DELIVERY', 'DELIVERED', 'CANCELLED'
limit	number	No	Maximum number of orders to return (default: 20)

Example Gemini Prompt:

"Gemini, list all pending orders that need packing right now."

Result Output:

Returns list of 5 pending orders with customer addresses and item lists.

2.4 search_customer_orders

Searches customer order history, past delivery addresses, and lifetime spending by customer phone number or name.

Parameter	Type	Req	Description
query	string	Yes	Customer mobile number (e.g. '9704173515') or name
Example Gemini Prompt:	"Gemini, search order history and total spend for customer phone 9704173515."		
Result Output:	Found 8 past orders for Sunny. Lifetime spend: ■14,200. Preferred address: Main Road, Rajam.		

2.5 update_order_status

Advances an order along the fulfillment pipeline (PENDING -> PREPARING -> OUT_FOR_DELIVERY -> DELIVERED -> CANCELLED).

Parameter	Type	Req	Description
orderId	string	Yes	The order ID to update
newStatus	string	Yes	One of: 'PENDING', 'PREPARING', 'OUT_FOR_DELIVERY', 'DELIVERED', 'CANCELLED'
Example Gemini Prompt:	"Gemini, mark order G-1786896453001-823 as OUT_FOR_DELIVERY."		
Result Output:	Order status updated to OUT_FOR_DELIVERY in AWS AppSync.		

3. Sales Analytics & Intelligence (4 Tools)

Provides financial intelligence, date-range revenue analytics, best-selling product rankings, and CSV export for accounting.

3.1 get_sales_report

Generates a detailed sales and revenue report for any custom date range (e.g. today, this week, or August 2026). Computes gross revenue, delivered/pending/cancelled counts, average order value, daily breakdown, and top-selling products.

Parameter	Type	Req	Description
startDate	string	No	Start date (YYYY-MM-DD, e.g. '2026-08-01')
endDate	string	No	End date (YYYY-MM-DD, e.g. '2026-08-29')
category	string	No	Optional category filter

Example Gemini Prompt:

"Gemini, generate a sales report for this month from 2026-08-01 to 2026-08-29."

Result Output:

Total Gross Revenue: █1,48,500 across 94 orders. AOV: █1,580. Top product: Lalitha HMT 26kg (42 bags sold).

3.2 export_orders_csv

Exports orders within a date range to structured CSV format suitable for opening in Microsoft Excel or Tally for accounting and GST audit.

Parameter	Type	Req	Description
startDate	string	No	Start date filter (YYYY-MM-DD)
endDate	string	No	End date filter (YYYY-MM-DD)
status	string	No	Status filter: 'DELIVERED', 'PENDING', 'ALL'

Example Gemini Prompt:

"Gemini, export all delivered orders for this month to a CSV file for my accountant."

Result Output:

Returns CSV data: OrderID, Date, CustomerName, Phone, Status, ItemCount, Subtotal, DeliveryFee, Total.

3.3 get_best_selling_products

Ranks top products in G-Store by total bags/units sold and revenue generated from completed orders.

Parameter	Type	Req	Description
limit	number	No	Number of top products to return (default: 5)

Example Gemini Prompt:

"Gemini, what are our top 5 best-selling products this month?"

Result Output:

1. Lalitha HMT Rice (█60,900), 2. Sona Masoori Rice (█42,000), 3. Freedom Oil (█18,400).

3.4 get_store_metrics

Computes high-level lifetime store analytics: total orders placed, gross lifetime revenue, pending orders, and recent order volume.

Example Gemini Prompt:	"Gemini, give me a quick summary of overall store metrics."
Result Output:	Lifetime Revenue: █4,82,000 Total Orders: 312 Pending Orders: 4 Average Order: █1,545.

4. Store Operations & Availability (3 Tools)

Provides real-time operational control to pause orders during heavy rains, update delivery radius, and manage minimum order amounts.

4.1 set_store_availability

Pauses or resumes incoming customer app orders by setting the store to Open or Closed with a public notice (e.g. heavy rain in Rajam, power cut, festival holiday, inventory counting).

Parameter	Type	Req	Description
isOpen	boolean	Yes	Set true to accept orders, false to pause orders
closingReason	string	No	Notice shown to customers (e.g. 'Heavy rain in Rajam — deliveries paused until 4 PM')
Example Gemini Prompt:	"Gemini, it is raining heavily in Rajam right now. Please close store orders with reason: Heavy rain in Rajam — delivery paused until 4:30 PM."		
Result Output:	Store status set to CLOSED in AppSync. Customer app displays the notice banner.		

4.2 get_store_settings

Reads active store configuration parameters from AWS AppSync including delivery radius (km), minimum order value (₹), and store open/closed status.

Example Gemini Prompt:	"Gemini, what are our current delivery radius and minimum order settings?"
Result Output:	Minimum Order: ₹150 Delivery Radius: 10.0 km Status: OPEN.

4.3 update_store_settings

Configures store delivery radius (in km), minimum order amount (in ₹), or store availability status.

Parameter	Type	Req	Description
minimumOrderAmount	number	No	New minimum order value in ₹ (e.g. 200)
deliveryRadiusKm	number	No	New delivery radius in km (e.g. 12.5)
isStoreOpen	boolean	No	Store open/closed boolean
closingReason	string	No	Reason text if closed
Example Gemini Prompt:	"Gemini, update store settings: set minimum order amount to ₹200 and delivery radius to 12 km."		
Result Output:	Store settings updated in sys_config: Min order ₹200, Radius 12 km.		

5. Daily Operational Prompt Cheat Sheet

Scenario / Task	Recommended Gemini Voice / Chat Prompt	Primary MCP Tool
Wholesale Truck Arrival	"Add 50 bags stock to Lalitha HMT 26kg and 30 bags to BPT Rice 26kg"	bulk_update_stock_or_prices
Add New Pack Size	"Add 10kg variant at ₹580 to Sona Masoori Rice with 40 bags stock"	add_product_variant
Generate Studio Photo	"Generate clean studio 600x600 packaging photo for Lalitha Rice 26kg bag"	generate_and_upload_product_image
Phone / Walk-in Order	"Create order for Ramesh (9876543210) for 1 bag Lalitha Rice and 2 Freedom Oils"	create_order
WhatsApp Receipt	"Generate WhatsApp bill for order G-1786896453001-823"	generate_order_invoice
Heavy Rain / Pause	"Close store orders: Heavy rain in Rajam — delivery paused until 4 PM"	set_store_availability
Monthly Sales Report	"Give me sales report from 2026-08-01 to 2026-08-29 with daily breakdown"	get_sales_report
Excel CSV Export	"Export this month's delivered orders to CSV for accounting"	export_orders_csv